Prohibitive to solve exactly

Barycenter Computation

- The OT barycenter problem is still an LP:

$$\min_{\substack{a \in \Delta_n, \\ (\pi_s \in \mathbb{R}^{n \times n_s})_s}} \left\{ \sum_{s=1}^{S} \lambda_s \langle \pi_s, C_s \rangle \;:\; \forall s,\; \pi_s \mathbf{1}_{n_s} = \mathbf{a},\; \pi_s^\top \mathbf{1}_n = \mathbf{b}_s \right\}$$

- But it is massive! **Prohibitive to solve exactly** as LP in most ML cases
- Instead, first order methods (e.g. sub gradient descent on the dual)

Sinkhorn Algorithm for Barycenters